Tiago Ferrão-Lindqvist

Data Engineering | Test & Validation Engineering

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PROFESSIONAL SUMMARY

Mechanical engineer with 12 years of experience in product design, testing and validation, and practical data tooling. Develops Python scripts to process test-rig data, check results against specifications and generate reports, including a nightly tool used by four test engineers. Experienced querying SQL Server test data and building Power BI dashboards for engineering users. Currently developing data engineering knowledge through an online course and personal projects.

PROFESSIONAL EXPERIENCE

Test & Validation Engineer | Corvane Automotive Components | Setúbal *Mar 2019–Present*

- Plan and run durability, vibration and climatic tests on seat-adjuster mechanisms across six servo-hydraulic and electric test rigs; prepare test reports for OEM customers.
- Develop Python scripts using pandas, NumPy, Matplotlib and npTDMS to process TDMS and CSV rig files, flag channels outside test specifications and generate report figures. The tool processes approximately 2 GB of data per week and reduced report preparation from about three days to half a day per test.
- Run the reporting tool nightly on a lab PC using Windows Task Scheduler. It writes summary results to an IT-provisioned staging table in the plant's SQL Server database and is used by four other test engineers.
- Query test results in SQL Server for trend analysis across three product generations.
- Built a Power BI dashboard using the staging table to show pass/fail trends and cycles to failure by product generation; used by about 15 validation and design engineers.
- Correlate test-rig results with FEA in ANSYS Mechanical and lead 8D root-cause analysis on field failures. Proposed a redesigned adjuster rail bracket that reduced warranty returns for that part by 40%.
- Maintain MATLAB/Simulink models of the adjuster drive used to select rig load profiles that match customer duty cycles.
- Trained three apprentices in rig safety and test setup (2022–2024).

Mechanical Design Engineer | Halvik Pumps | Porto *Sep 2014–Feb 2019*

- Designed industrial centrifugal-pump casings and impellers in SolidWorks and prepared production drawings with GD&T; owned DFMEA for two product lines.
- Performed structural FEA of pump casings in ANSYS. Supported the change of one casing family from grey cast iron to ductile iron, saving 12% in material cost.
- Supported conversion of the pump performance test bench to a PLC-controlled setup, including changes to setpoints and timers in Siemens TIA Portal alongside the automation engineer.
- Wrote a LabVIEW data-acquisition program for the performance test bench, reducing test setup time from about two hours to 30 minutes.

SELECTED DATA PROJECTS

Bike-sharing Trips API *Personal Learning Project | 2025*

- Built a small Flask API over a public bike-sharing dataset loaded into SQLite, with three read-only endpoints.
- Followed a tutorial to run the project using Docker Compose; maintained personal repositories with Git.

Rig Data Processing and Reporting Tool *Corvane Automotive Components | 2021–Present*

- Built Python scripts to parse test-rig output, validate channels against test specifications and generate standard report figures.
- Technologies: Python, pandas, NumPy, Matplotlib, npTDMS, SQL Server, Windows Task Scheduler.

Test Results Dashboard *Corvane Automotive Components | 2023*

- Created a Power BI dashboard showing pass/fail trends and cycles to failure per product generation for validation and design engineers.
- Technologies: Power BI, SQL.

Adjuster Drive Load-Profile Model *Corvane Automotive Components | 2020–Present*

- Maintains a MATLAB/Simulink model of the seat-adjuster drive, including motor, gearbox and rail friction, to help select rig load profiles.
- Technologies: MATLAB, Simulink.

EDUCATION

MSc Mechanical Engineering | Sado Valley Polytechnic *2014*
Thesis: Fatigue life prediction of welded brackets under variable-amplitude loading

PROFESSIONAL DEVELOPMENT

Online Data Engineering Course *In progress*
Completed 6 of 9 modules.

TECHNICAL SKILLS

Data and programming: Python (pandas, NumPy, Matplotlib, npTDMS), SQL, MATLAB/Simulink, Flask, SQLite, Excel/VBA

Data reporting and analysis: Power BI, LabVIEW, Minitab, design of experiments (DOE)

Engineering: ANSYS Mechanical, SolidWorks, CATIA V5, GD&T, DFMEA, 8D problem solving

Tools: Git (personal projects), Docker Compose (personal learning project), Windows Task Scheduler

LANGUAGES

Portuguese (native) | English (fluent) | Spanish (good)